Rare and higher-order Solinas point patterns

2026-10-12 · higher-order-20261012 · measured descriptive observations

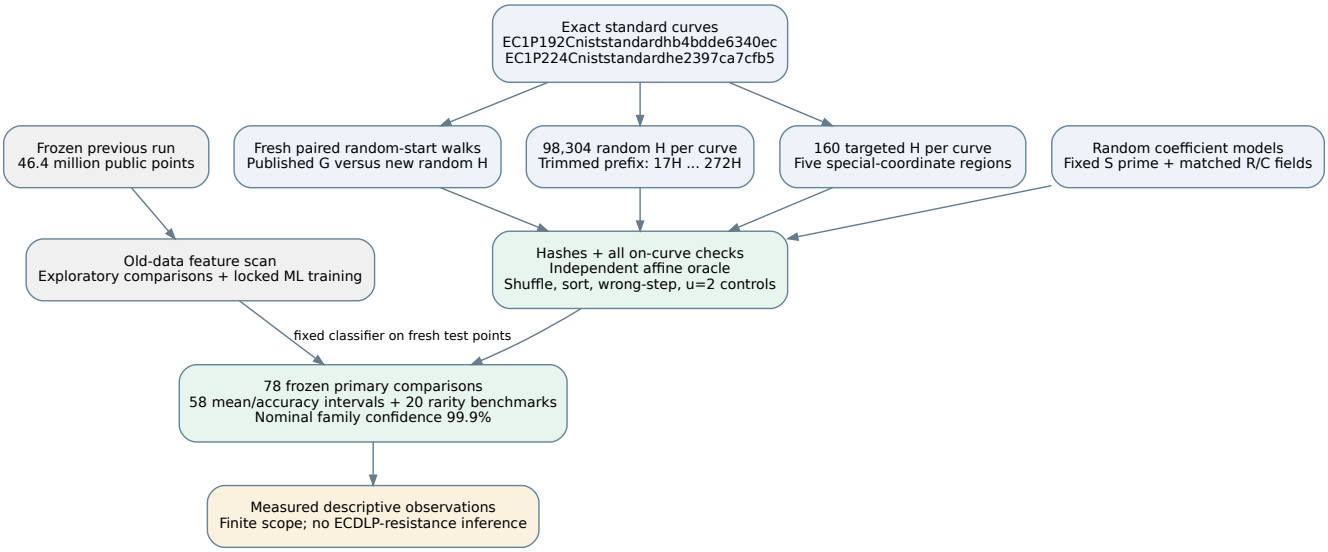
Measured: 0 of 58 primary mean/accuracy intervals exclude zero; 0 of 20 published-prefix reference tests cross the corrected threshold. The tested features did not resolve a published-generator or fixed-S-prime effect in this fresh batch.

Scope: 77.4 million fresh public points; two exact standard curves; one published G per curve; matched random generators and random-field models. Page 6 adds 6.29 million fresh points for three selected rare-prefix candidates. No independent scientific review, official hypothesis transition or relative ECDLP-resistance result. Full values and provenance: report.md, comparisons.csv, analysis.json and verification.json next to the portable task.

A higher-order relation did appear — the ordinary curve law

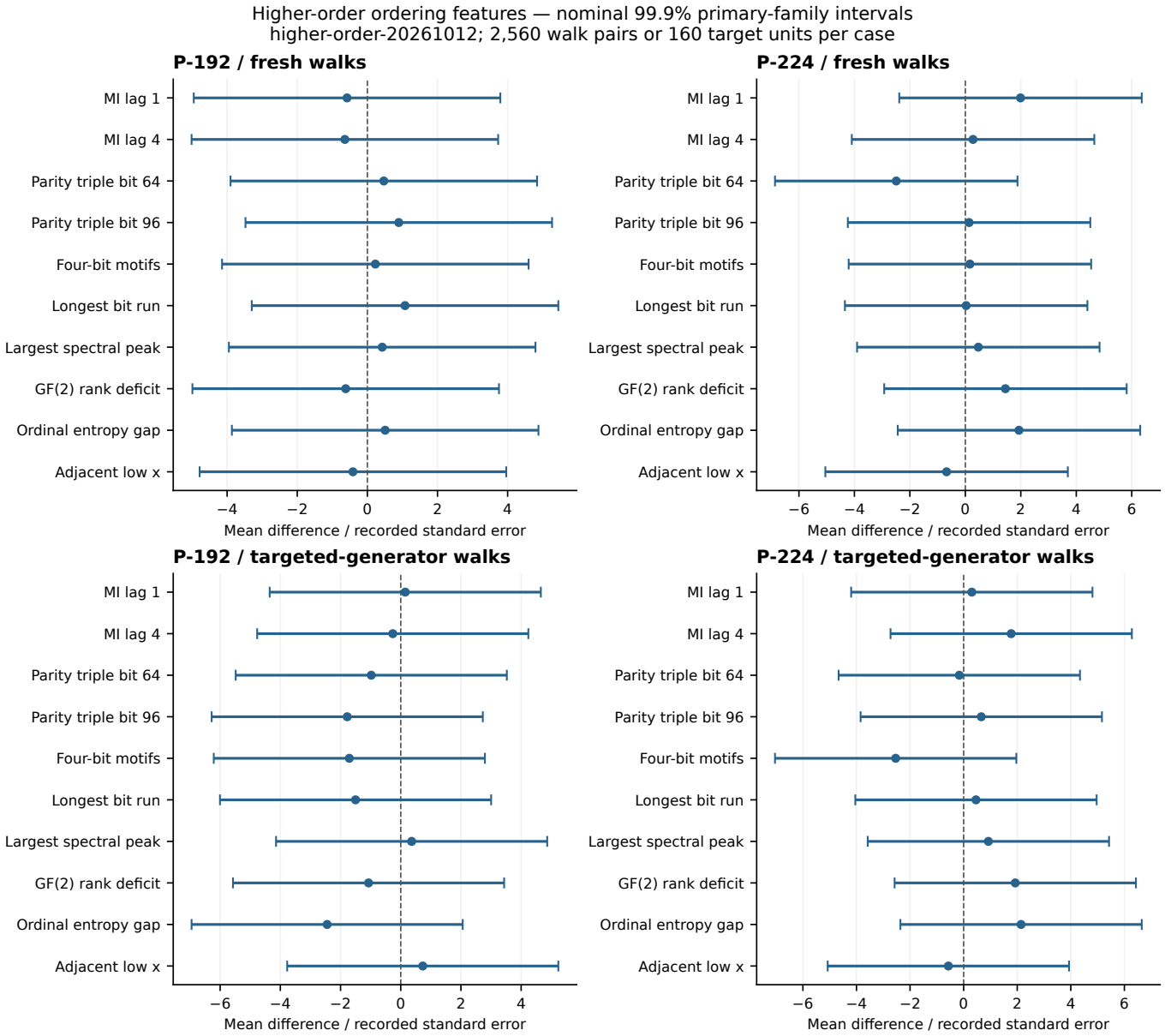
For a generator with x coordinate u and consecutive points with x values L,X,R, `(L+R)(X−u)^2 = 2((Xu+A)(X+u)+2B) mod p`. Add the formulas for x(Q−H),x(Q+H), then substitute $y^2=x^3+Ax+B$. This exact identity is derived for every nonsingular short Weierstrass curve with affine neighbors. It is not specific to a Solinas prime and reveals no unknown scalar.

Measured: 1,677,840 spread triples satisfied it; wrong step 0/2048, shuffled 0/2048 hits. Low simple correlations never imply an absence of algebraic structure.



Fresh ordered-walk and targeted-generator comparisons

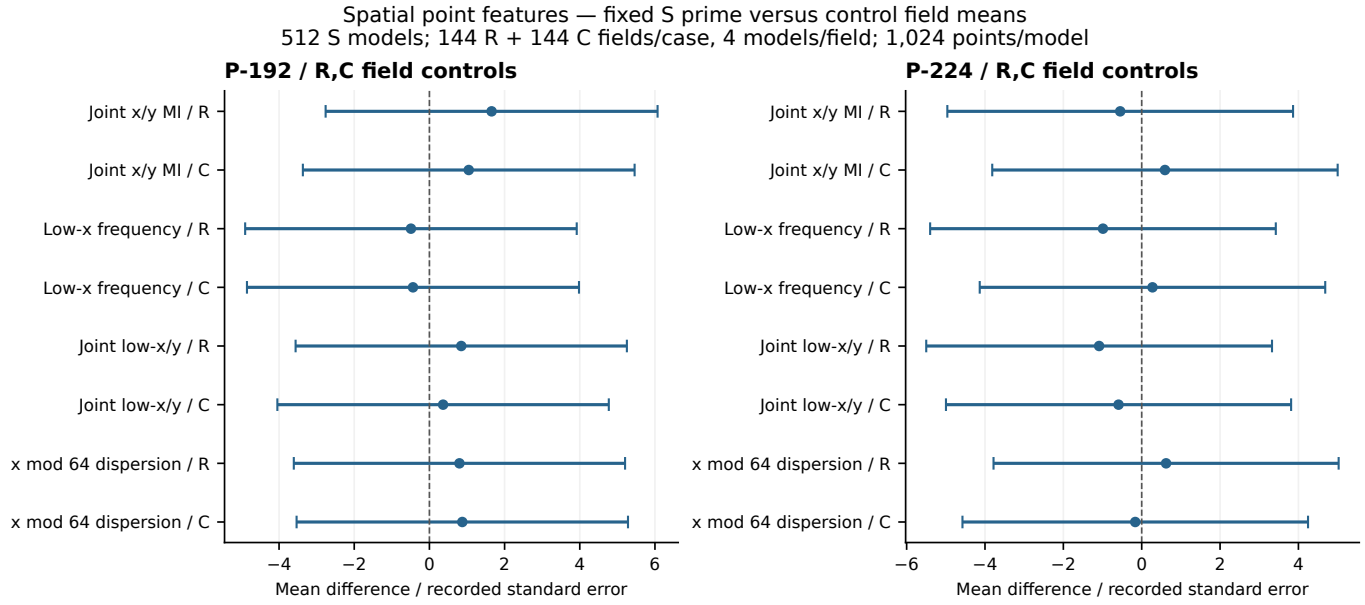
2,560 paired independent-start blocks ×2,048 points/cohort/case. 160 selected generators/case near x=0,2^64,2^96,p−2^64,p−1; four 1,024-point starts averaged per generator. Targets are a finite chosen set, not a uniform sample. All prefix comparisons use multiples 17 through 272, excluding the first 16 multiples. Larger prefix scores measure the declared anomaly tails.



Nominal 99.9% confidence for the 78-comparison primary family: Bonferroni α/test=1.282051e-05. 58 mean/accuracy intervals use Student/Welch approximations; 20 reference tests use random-generator ranks. Mean plot units = difference / recorded standard error. Original units and all values are in comparisons.csv. Binary Hankel rank is a sequence matrix statistic, not group rank or ECDLP complexity. Companion controls are not independent samples.

Fresh spatial point comparisons and ML holdout

3,328 existing frozen random models; 578 distinct primes. Per case: 512 models on fixed S p, 144 R plus 144 C fields, four models/field, 1,024 uniform affine points/model. R matches field size; C also matches p mod 12. Random control models are averaged by field. New points are independent of the old scan; field/model draws themselves are reused, not new replicates.



Feature-only ML, evaluated once on fresh seed blocks

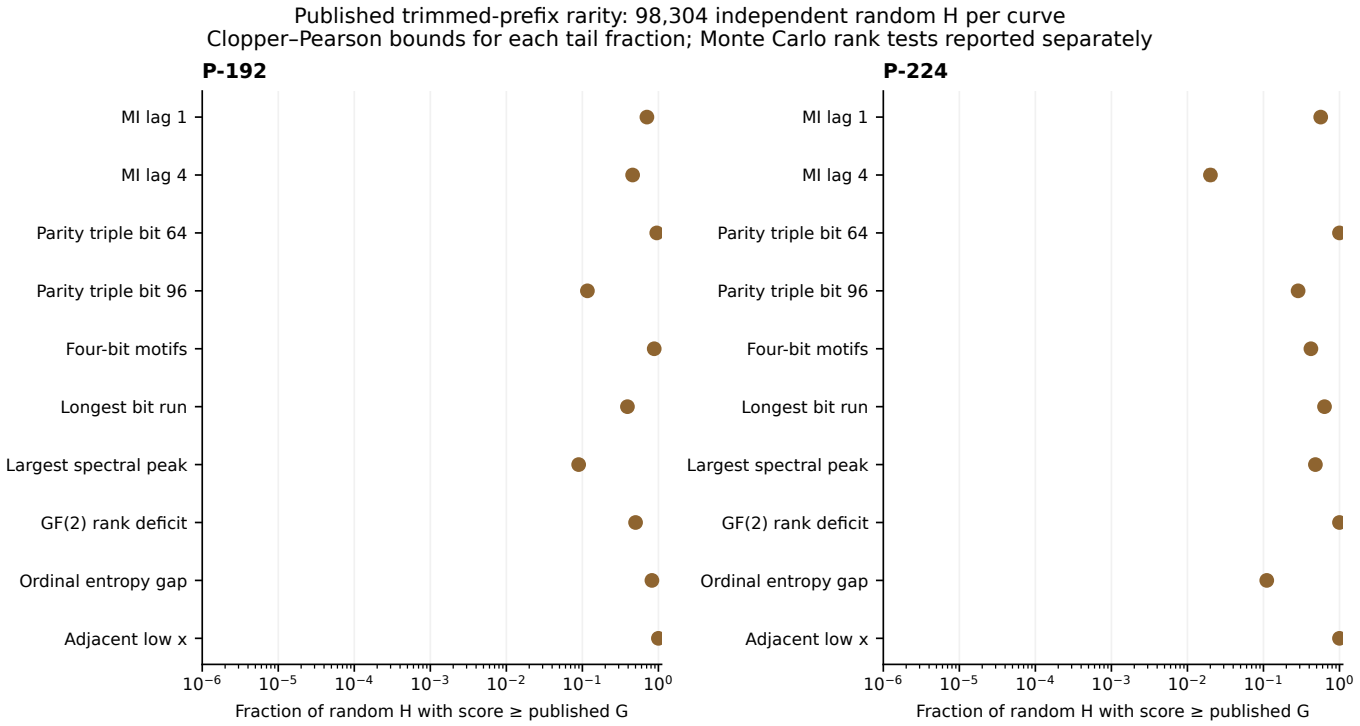
Train only on old walks trimmed to 2,048 points; test on new 2,048-point walks. Algorithm and settings were fixed before scanning. Input = ten numerical descriptors, with no G coordinates, IDs, seed, scalar or label. Pair correctness values before interval estimation; no per-point inference.

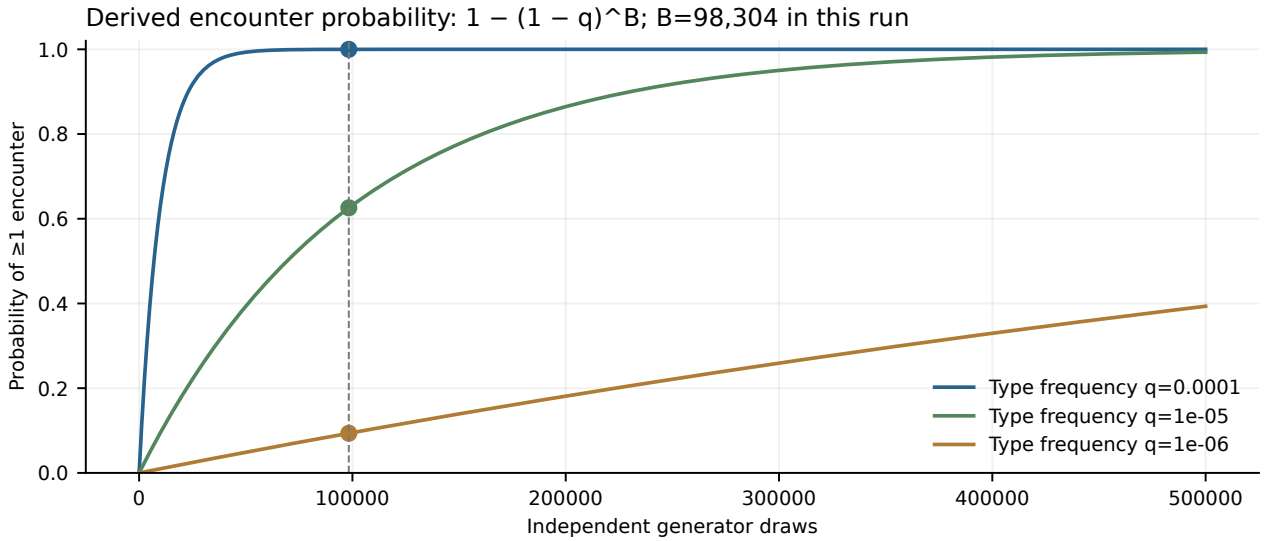
Case	Accuracy	Corrected approximate interval	Pairs
P-192	49.4141%	46.3903% to 52.4379%	2560
P-224	49.7461%	46.6800% to 52.8122%	2560

Old scan: 46,399,488 points, 56 exploratory comparisons; 0 zero-excluding intervals. Fresh primary results are reported separately. Secondary region intervals (100-test exploratory family): 0 exclude zero. All observations, favorable or otherwise, remain in analysis.json. Uniform nonzero multiples have exactly identical distributions for every generator; only ordering/window effects could differ.

Published-prefix rarity and sampling limits

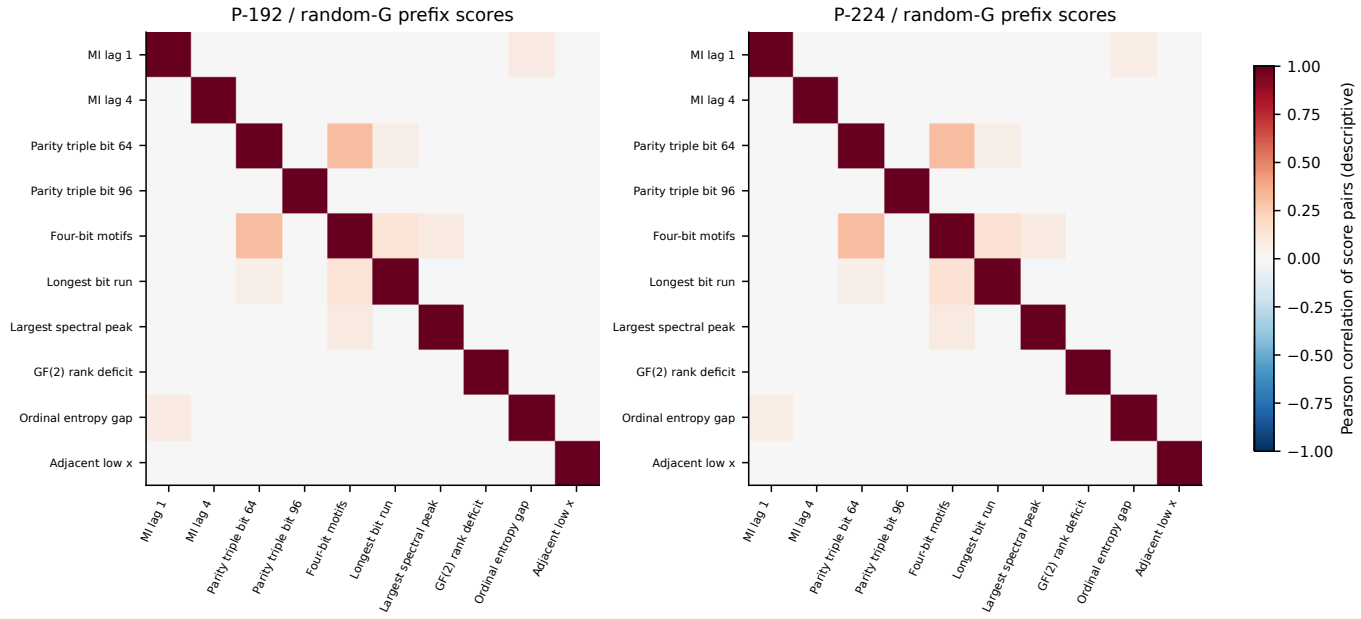
The random-generator bank has 98,304 independent H per exact standard curve. Reference p = (1 + count with score ≥ official)/(98,304+1), ties included; minimum p≈1.017×10^−5. 0 of 20 tests cross the corrected threshold. This is a uniform-G benchmark, not a claim about historical G selection.





Encounter probabilities: q=10^−4 → 99.9946%; q=10^−5 → 62.5830%; q=10^−6 → 9.3627%. Encountering a type does not guarantee recognition by these features. Types with density near 1/p need deliberate targeting. Point count does not imply proportional coverage of a 192-/224-bit field.

Joint score behavior, integrity and reproducibility



Exploratory: these descriptors share bits/points. Correlations and joint tail counts (bank-joint-tails.csv) describe that dependence. Empirical 99th thresholds use the same bank; no independence or prime-specific effect is inferred. Selected bank maxima are expected order statistics in a large search.

Measured integrity: 77,414,912 points on-curve; all raw SHA-256 matched; 213,058 feature units replayed. Separate integer EC oracle: 1,677,840 transitions, 5,314 scalar/start checks. All sorted positive controls passed; 606,720 isomorphic u=2 points checked. Same-session verification is not an independent research review.

Sampling: 819.92 s wall, 2383.03 summed job CPU s, three workers, 11,842 jobs, zero failures. Frozen source: `29f2459899a4b8c5a6581a459132bb4d1103a582`. Exact standard aliases: EC1P192Cniststandardhb4bdde6340ec and EC1P224Cniststandardhe2397ca7cfb5. Full UIDs, p,A,B,n,G and internal certificate provenance are in inputs.json and report.md. Environment/command/dirty-state metadata and content hashes are retained.

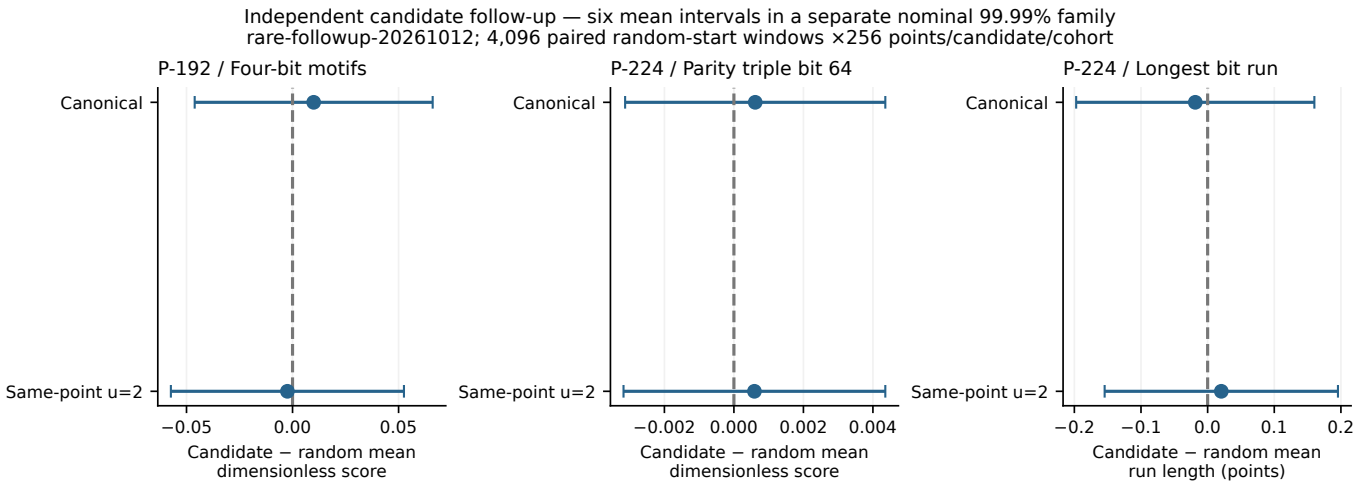
Agent-free reruns: use research.py batch with a new public seed and a new output directory. The dataset includes old discovery features for –reuse-discovery; no old raw rescan is needed. Keep coordinate/JSONL shards and manifests for future analyses; pool batches only under a prospective plan.

No relative ECDLP hardness, recovery, security exponent or attack-cost bound is measured. Finite feature scope and rare families remain open beyond the declared observations. Historical reports/frontier graphs remain unchanged.

Three rare-prefix candidates: independent fresh follow-up

Three of 3,200 targeted descriptors fell in a tail containing at most 0.1% of the random-G bank. Strongest: P224 x(H)=2^64+2, bit-64 run length 22 in its trimmed prefix (3/98,304 bank generators as extreme); same-point u=2 changes that score to 13. These are exploratory selected outliers.

New protocol frozen and pushed before its distinct public seed: 4,096 paired random-start windows ×256 points per candidate/cohort, fresh random control G per pair. 6,291,456 additional points, all verified; total in both scientific stages: 83,706,368.



Measured: 0 of six mean intervals excludes zero; 0 of three exact paired event-rate tests crosses the corrected threshold. This is a separate nominal 99.99% nine-test family (α=.0001), alongside the primary 99.9% family. Do not combine them into a claimed 87-test 99.9% family; simple union coverage is nominal 99.89%, with approximate mean-interval assumptions.

Original rare score	Threshold	Target hits	Random hits
P192 motifs	4.2685	2/4096	1/4096
P224 triple parity	0.2126	7/4096	5/4096
P224 longest run	22	0/4096	0/4096

The exact paired p values are 1, 0.7744 and 1. Zero 22-bit runs leave a corrected arm-rate upper bound of 0.285%; very rare effects remain possible. The earlier deterministic prefix remains reproducible. Fresh tests concern random starts, not erasing a selected scalar-window observation. Same-point u=2 is a coordinate control, not an independent curve.

All 6,291,456 points/hash/score replays passed; independent integer oracle checked 196,608 transitions, 1,536 starts and 384 control generators. Wall 55.91s; CPU sum 164.82s; zero job failures. Full raw mean/rate bounds and candidate G are in report.md and follow-up analysis.json. Ordinary-prime matched candidates would be needed for prime-specific inference.